

ORIGINAL RESEARCH ARTICLE

Sotatercept Improves Right Heart Mechanics Despite Reduced Myocardial Contractility in Pulmonary Arterial Hypertension: A Prospective Exercise Hemodynamic Study

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BACKGROUND: The right ventricular (RV) response to pressure overload is a crucial determinant of outcomes in pulmonary arterial (PA) hypertension. Sotatercept improves clinical outcomes, but its effects on the right ventricle have been inconsistent. We evaluated the effects of sotatercept on load-independent measures of RV contractility, RV-PA coupling, and right heart deformation at rest and exertion in patients with PA hypertension.

METHODS: In this prospective trial among participants with PA hypertension recruited between July 2024 and May 2025, we performed simultaneous echocardiography and invasive pressure measurements during rest and exercise to evaluate changes in RV contractility (end-systolic elastance [Ees]), afterload (PA elastance [Ea]), RV-PA coupling (ratio of arterial Ea to Ees), and right heart deformation, including RV and right atrial strain, before and after 24 weeks of sotatercept using mixed models.

RESULTS: Among 30 participants (n=21; 70% female; mean age, 49.3 years [SD 13.5]), sotatercept decreased PA Ea. These changes were associated with reduced RV contractility, PA pulsatility index, and the maximal rate of RV pressure increase over time (dP/dt_{max} ; $P<0.001$ for all). The decrease in Ees was associated with a decrease in PA Ea ($r=+0.70$ [95% CI, +0.44 to +0.85]; $P<0.0001$) with preserved RV-PA coupling (mean difference, +0.08 [95% CI, -0.11 to +0.26]; $P=0.42$). Although systolic indices measured at the base of the right ventricle (tricuspid annular plane systolic excursion and systolic velocity of the lateral tricuspid annulus) decreased after sotatercept use, there was an increase in RV free wall-dependent systolic indices, including resting and exercise RV free wall strain (mean difference, +6.2% [95% CI, +4.5 to +7.8]; $P<0.0001$) and fractional area change ($P=0.0001$). Beyond the right ventricle, there was also improved right atrial filling and contractile performance across rest and exercise ($P<0.0001$ for all).

CONCLUSIONS: Sotatercept decreases RV contractility in proportion to reduced afterload, while preserving the RV-PA coupling relationship. Despite decreased RV contractility, global measures of right heart performance, such as RV free wall strain and RA strain, improve, alongside redistribution of RV deformation from annular motion to free wall shortening. These data suggest that sotatercept causes an energetically favorable reduction in resting RV contractility to match the lower afterload, which does not appear physiologically detrimental when considering the concurrent improvement in RV free wall contractile mechanics and exertional right heart reserve.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06409026.

Key Words: exercise ■ heart atria ■ pulmonary arterial hypertension

Until recently, pulmonary vasodilator therapy was the primary mode of treatment for pulmonary arterial hypertension (PAH), with vasodila-

tion favorably lowering right ventricular (RV) afterload, thereby improving RV systolic performance.¹ In contrast to vasodilators, sotatercept is a novel PAH therapy that

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Clinical Perspective

What Is New?

- Sotatercept decreased right ventricular (RV) contractility in pulmonary arterial hypertension when assessed using load-independent measurement of end-systolic elastance.
- The reduction in RV contractility occurs proportional to the reduction in afterload with preserved ventricular–vascular coupling.
- Despite reduction in contractility, integrated measures of right heart performance using echocardiographic strain assessment of RV free wall longitudinal motion and right atrial strain improve after sotatercept treatment.

What Are the Clinical Implications?

- Although RV contractility decreases after sotatercept treatment, this may represent a physiologically appropriate reduction to match sotatercept-induced afterload reduction, with preservation of the ventricular–vascular coupling relationship.
- Consistent with the reduction in contractility, measures of basal longitudinal motion, such as tricuspid annular plane systolic excursion or systolic velocity of the tricuspid annulus, may decrease after sotatercept use, whereas measures of global chamber performance, including RV free wall strain or right atrial strain, more reliably characterize salutary effects on the right heart after sotatercept treatment.

is proposed to alter proliferative signaling in the pulmonary vasculature through activin modulation, thereby lowering RV afterload through predominantly nonvasodilatory mechanisms. Both vasodilators and sotatercept cause afterload reduction; however, pulmonary vasodilator therapy has consistently improved RV systolic function,¹ whereas sotatercept has generally not improved most RV systolic indices. Indeed, some studies even suggested that sotatercept may decrease RV systolic performance and cardiac output despite improved exercise performance.^{2–5} Thus, uncertainty remains regarding the potential for a direct negative inotropic effect of sotatercept on RV contractility, with unclear implications for myocardial reserve during exercise and associated exercise capacity.

As opposed to traditional load-dependent echocardiographic assessments of RV systolic function, load-independent measures, such as end-systolic elastance (Ees), more directly reflect underlying RV contractility, particularly when interpreted in the context of global RV afterload (pulmonary arterial [PA] elastance [Ea]). Beyond assessing Ees and Ea as individual measures, interpreting their direct relationship (Ea/Ees) can describe RV contractile performance relative to afterload, which is referred to as RV–PA coupling under normal physiologic conditions.

Nonstandard Abbreviations and Acronyms

Ea	effective arterial elastance
Ees	end-systolic elastance
FAC	fractional area change
PA	pulmonary arterial
PAH	pulmonary arterial hypertension
PAP	pulmonary arterial pressure
RA	right atrial
RHC	right heart catheterization
RV	right ventricular
RVEDA	right ventricular end-diastolic area
RVEDP	right ventricular end-diastolic pressure
RVFWS	right ventricular free wall strain
RVS	right ventricular stroke work
TAPSE	tricuspid annular plane systolic excursion

A decrease in Ees relative to Ea suggests RV–PA uncoupling, in which RV contractile augmentation fails to match a heightened afterload. In addition, resting measures of RV contractile performance do not adequately reflect the dynamic range of RV systolic augmentation reserve that is provoked by exertional stress when symptoms occur.

The effects of long-term sotatercept therapy on load-independent RV contractility, standard RV–PA coupling indices, and RV systolic performance during exertion remain unknown, forming the primary basis for this analysis. To provide comprehensive insight into global right heart performance, we also measured RV free wall strain (RVFWS), right atrial (RA) strain,⁶ RA operating compliance, and RA contractile function, along with multiple invasive, noninvasive, and integrated indices of RV systolic performance during rest and exercise. We hypothesized that RV contractility would decrease in patients with PAH after sotatercept treatment, as shown previously,⁵ but that this would represent an appropriate decrease in contractility to match the lower afterload, with preserved RV–PA coupling. We postulated further that in contrast to RV contractility indices, global measures of right-sided chamber performance as assessed by the deformation indices of RVFWS and RA strain during rest and exercise would improve with sotatercept treatment. In this prospective study, we tested these hypotheses through exercise echocardiography and micromanometer right heart catheterization (RHC) before and after 24 weeks of sotatercept therapy, quantifying rest and exercise right heart mechanics, Ees, and Ea/Ees.²

METHODS

Study Overview

This was a single-arm, prospective, nonrandomized study in which all participants underwent exercise hemodynamic

evaluation before and after 24 weeks of sotatercept therapy. The data that support the findings of this study are available from the corresponding author upon reasonable request. All participants provided written informed consent and the study was approved by the Mayo Clinic Institutional Review Board (24-002570).

Participants had group 1 PAH and were symptomatic (New York Heart Association Functional Class $\geq$ II) despite existing PAH therapy, with planned initiation of sotatercept at the Mayo Clinic (Rochester, MN). Patients were required to demonstrate persistent pulmonary hypertension (mean PA pressure [PAP] >20 mm Hg with pulmonary vascular resistance >2 Wood units) on existing therapy at the time of recruiting RHC. Full inclusion and exclusion criteria are included in the [Supplemental Material](#). Funding was provided by Merck for this investigator-initiated trial with a contract that provided no limitations on data to be considered for publication.

Invasive Hemodynamic Exercise Testing

All participants underwent invasive hemodynamic exercise testing during supine cycle ergometry as described previously.⁷⁸ This included RHC through a 9-Fr internal jugular sheath with a 4-Fr radial arterial cannula for blood pressure monitoring, blood gas sampling, and direct Fick cardiac output calculation, and a 5-Fr femoral venous infusion catheter for femoral venous blood gas sampling. All right heart tracings including the RV waveform were recorded using high-fidelity, solid-state micromanometers digitized at 240 Hz for blinded offline analysis. RA pressure was recorded continuously throughout rest and exercise through the side arm of the internal jugular sheath. Following resting phase assessment, exercise was initiated at a common starting workload of 20 W (60 rpm) for 2 minutes, followed by an increase in exercise stage by 20 W every 2 minutes until peak exertion. The primary results describing hemodynamic changes, peripheral oxygen transport, and surrogate estimates of RV–PA coupling using standard echocardiographic measures of RV performance (tricuspid annular plane systolic excursion [TAPSE], RV s' velocity, and fractional area change [FAC]) adjusted for PA systolic pressure have been reported.² The current analyses focused on detailed RV waveform analyses with single-beat Ees (load-independent contractility), pulmonary vascular load (PA Ea), and ventricular–vascular coupling relationship (Ea/Ees) during rest, matched-workload 20-W exercise, and peak exercise. We also evaluated echocardiographic RV and RA strain, RA volumes, and RA volumetric-based function during rest and exercise. All hemodynamic data including echocardiography and RV waveforms were interpreted in a blinded fashion regarding timing at baseline or after sotatercept treatment.

Given the highly invasive nature of the protocol, with radial, internal jugular, and femoral venous access over 2 study visits, and the presence of 4 randomized placebo-controlled trials demonstrating consistent efficacy for sotatercept in improving symptoms and death or transplantation outcomes in PAH,^{9–12} it was considered unethical to include a placebo group. The study was therefore designed as a single-arm, open-label treatment trial (Sotatercept on Central Cardiopulmonary Performance and Peripheral Oxygen Transport During Exercise in Pulmonary Arterial Hypertension; URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06409026) in which all patients were

provided sotatercept after baseline invasive hemodynamic evaluation, with repeat invasive evaluation at 24 weeks.

Catheterization Assessment of RV Performance

High-fidelity right heart pressures were measured in a blinded fashion at end-expiration during rest and exercise. Pulmonary artery pulsatility index was calculated as (PA systolic–PA diastolic pressure)/RA pressure, and has been shown to correlate with RV sarcomeric contractility as a measure of RV contractile performance.^{13,14} RV stroke work (RVSW) was calculated as stroke volume $\times 0.0136 \times (\text{mean PAP} - \text{RA pressure})$,¹⁵ and after correction for RV end-diastolic area (RVEDA) was used as a surrogate for preload-adjusted stroke work during rest and exercise (RVSW/RVEDA) as a measure of contractile performance.

PA effective elastance (Ea) was computed as the ratio of manually measured PA end-systolic pressure and direct Fick-derived stroke volume at rest and during exercise using simultaneously measured oxygen consumption. High-fidelity micromanometer RV pressure tracings were obtained at rest, immediately before PAP measurement. RV end-diastolic pressure (RVEDP) was measured at end-expiration, with RVEDP divided by concurrent echo RVEDA as a measure of resting RV operating stiffness. A representative RV pressure tracing at end-expiration was manually selected and exported at 4-ms resolution for single-beat analysis. Briefly, a sinusoidal wave was fit to the isovolumetric contraction and relaxation phases of the RV waveform as previously described⁵ to estimate the theoretical maximal RV systolic pressure (Pmax) that would have been generated with RV contraction against a closed pulmonary valve. Using Pmax, directly measured PA end-systolic pressure, and Fick stroke volume, RV Ees (a load-independent measure of RV contractility) was calculated as $(\text{Pmax} - \text{PA end-systolic pressure}) / \text{stroke volume}$. RV–PA coupling index was then computed as Ea/Ees describing the degree of matching between RV and pulmonary properties. In addition, the maximal rate of RV pressure rise over time (dP/dt_{max}) was quantified from the isovolumetric contraction waveform as an independent measure of RV contractile performance. To account for the load-dependency of dP/dt_{max} for RV contractile performance, dP/dt_{max} was also adjusted for instantaneous absolute pressure.

Due to the complex exercise protocol, which also required femoral venous blood gas sampling and femoral thermodilution with 2-minute exercise stages, RV pressure waveforms were only measured at rest. To estimate RV Ees during exercise, RV Ees was independently estimated using a general approach that uses a validated cardiovascular model with a parameter optimization algorithm, as previously validated.^{16,17} Briefly, invasive measurements of cardiac output, heart rate, central venous pressure, pulmonary capillary wedge pressure, arterial pressure, and pulmonary artery pressure are used as inputs to the model, along with echocardiographic ejection fraction, and in the current study we also incorporated measured RVEDA as an input variable to account for RV remodeling in PAH. The algorithm searches the multidimensional space composed of the model parameters to optimize the agreement of all hemodynamic parameter measurements with the model output. The algorithm is implemented in real-time simulation (<http://harvi.online>) with optimization of left ventricular and RV Ees in an iterative approach to minimize error for model fit for all

measured parameters.^{16,17} PA Ea during exercise was directly measured from end-systolic pressure and Fick stroke volume, with exercise Ea/Ees used to describe exertional ventricular-vascular coupling.

Echocardiographic Assessment of Right Heart Performance

RVFWS was measured from an RV-focused 4-chamber view using commercially available software (Image Arena; TomTec Imaging Systems). RA strain was calculated as the average of strain in 6 segments from the 4-chamber view to calculate reservoir, conduit, and contractile strain, reflecting atrial filling, passive atrial emptying, and atrial contractile performance, respectively (Figure S1).^{18,19} RA maximal volume, preatrial contraction volume, and minimal volume were calculated by summation of the discs in the 4-chamber view. RA ejection fraction was calculated as (maximum–minimum volume)/maximum volume. RA expansion index was calculated as (maximum–minimum volume)/minimum volume. TAPSE, RV s', and FAC were measured at rest and exercise as previously described.²⁰

Integrated Pressure–Volume Assessment of RA Performance

Detailed RA waveform analysis was used to comprehensively characterize phasic RA pressures at end-expiration throughout the cardiac cycle, including during atrial contraction (peak a wave), atrial relaxation (x nadir), atrial filling (peak v wave), and atrial emptying (y nadir). The RA distending pressure for RA filling was calculated as peak RA v wave–x nadir pressure. The pressure generated upon atrial contraction or RA booster pressure was calculated as peak RA a wave–y nadir pressure (Figure S1). By indexing the atrial deformation during atrial filling with associated pressure rise, we estimated RA capacitance as previously described (reservoir strain/distending pressure).¹⁸ Similarly, by indexing atrial deformation during atrial contraction with associated pressure rise, we estimated RA booster efficiency (booster strain/booster pressure; Figure S1).

Using simultaneous wedge pressure tracings, inspiration was identified by the decline in wedge pressure during which end-inspiratory mean RA pressure was measured. Because RA pressure should normally decline with inspiratory pleural pressure reduction in the presence of adequate RA capacitance to accommodate the increased venous return of inspiration,²¹ the degree of inspiratory RA pressure reduction (RA end expiration–RA end inspiration) during rest and exercise was also assessed as a surrogate for RA capacitance.

Statistical Analysis

Changes in all measures from baseline (pre-sotatercept) to 24 weeks (post-sotatercept) at each individual phase were assessed using linear mixed models with participant ID as a random effect and pre- versus post-sotatercept study phases as a fixed effect using a compound symmetry covariance structure with denominator degrees of freedom estimated by the Satterthwaite approximation. To model the collective change across all the 3 paired phases (rest, 20 W, and peak exercise) for key variables of interest, linear mixed models were fit with pre–post sotatercept timepoints, hemodynamic phase,

and their interaction (pre–post*hemodynamic phase) as fixed effects, with participant ID as a random effect. The interaction term tested the hypothesis of whether the treatment effect differed across hemodynamic phases. The primary end point of the trial of peak exercise mean PAP/cardiac output had not been previously reported for sotatercept in PAH, requiring assumptions for power calculations. We initially planned to enroll 21 participants, providing ≈97% power (2-sided $\alpha=0.05$) to detect a conservative mean within-subject reduction of 1.0 mm Hg·L·min, assuming an SD of within-subject change of 1.06 mm Hg·L·min derived from previous data in our laboratory.²² Due to faster than expected recruitment, the target sample size was subsequently increased to 30 participants to improve precision of effect estimates for secondary measures.

Post hoc Pearson correlation analyses were used to explore potential correlations between key variables of interest. Given the modest sample size and mechanistic nature of the investigation, no hierarchical testing structure was predefined for secondary end points, and multiple exploratory end points were evaluated. The *P* values provided are not corrected for multiplicity and should be considered interpretative aids as they no longer accurately represent the risk of false discovery. Missing values were not imputed due to the use of mixed models, which can flexibly handle missing data. The current report adhered to STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) reporting guidelines. All analyses were conducted using 2-sided tests with an α level of 0.05. Analyses were performed with JMP Pro, version 19.0.0 (JMP Statistical Discovery LLC) and R studio.

RESULTS

Baseline Characteristics

A total of 30 participants with PAH (21 women [70%]) were enrolled between July 2024 and May 2025 (Figure S2), with a mean age of 49.3±13.5 years.² Participants displayed severe pulmonary hypertension at rest, with an average mean PAP of 50±12 mm Hg and pulmonary vascular resistance of 8±4 Wood units. Most participants (*n*=23 [77%]) were on dual or triple PAH therapy at enrollment, and 5 (17%) were on parenteral prostaglandin therapy (Table 1). RV dysfunction prevalence varied by measurement technique, with 6 (21%) having an abnormal TAPSE, 9 (32%) abnormal RV s', 11 (41%) abnormal RVFWS, and 18 (72%) abnormal FAC. All participants received their first dose of sotatercept immediately after their initial qualifying RHC, with all participants returning for their second RHC at a median of 25 weeks (interquartile range, 23–26) weeks.

Effect of Sotatercept on RV Contractility and Ventricular–Vascular Coupling

As previously reported, sotatercept decreased mean PAP,²⁰ with a decrease in PA Ea (mean difference, –0.27 mm Hg/mL [95% CI, –0.43 to –0.11]; *P*=0.002). This was coupled with a significant decrease in

Table 1. Baseline Characteristics

Characteristics	Values
Age, y	49.3±13.5
Female sex	21 (70)
Body mass index, kg/m ²	28.0±5.4
Race	
White	27 (90)
Black	1 (3)
Asian	2 (7)
Comorbidities or pulmonary hypertension history	
Hypertension	3 (10)
Diabetes	3 (10)
Atrial fibrillation, paroxysmal or persistent	1 (3)/0
Previous cardiac surgery	1 (3)
Creatinine, mg/dL	1.0±0.3
NT-proBNP, pg/mL	228 (90–942)
Six-minute walk distance, m	439.5±113.4
Cause of pulmonary hypertension	
Idiopathic or hereditary	11 (37)
Connective tissue disease	8 (27)
Congenital heart disease	3 (10)
Drug or toxin	8 (27)
Medication	
Dual therapy	11 (37)
Triple therapy	12 (40)
Phosphodiesterase 5 inhibitor	22 (73)
Intravenous or subcutaneous prostaglandin	5 (17)
Loop diuretic	18 (60)
Right heart dysfunction	
Fractional area change <35	18 (72)
TAPSE <17 mm	6 (21)
RV s' velocity <9.5 cm/s	9 (32)
RV free wall strain <20%	11 (41)

Values represent mean±SD, n (%), or median (interquartile range). Total n=30 patients with pulmonary arterial hypertension. NT-proBNP indicates N-terminal pro-B-type natriuretic peptide; RV, right ventricular; and TAPSE, tricuspid annular plane systolic excursion.

micromanometer-measured RV P_{max} (mean difference, −27.7 mmHg [95% CI, −36.3 to −19.0]; $P<0.0001$) and $dP/dt_{\max}$ (mean difference, −124 mmHg/s [95% CI, −170 to −77]; $P<0.0001$), along with decreased RV Ees (mean difference, −0.29 mmHg/mL [95% CI, −0.42 to −0.17]; $P<0.0001$), consistent with reduction in resting contractility (Figure 1 and Table 2). With balanced decreases in Ea and Ees, there was no statistically significant change in Ea/Ees (mean difference, +0.08 [95% CI, −0.11 to +0.26], $P=0.42$). Similar to resting changes in isovolumetric pressure-measured Ees, model-estimated Ees also decreased comparably after sotatercept treatment (mean difference, −0.39 [95%

CI, −0.58 to −0.20]; $P=0.0002$). There was no significant difference or systematic bias when comparing the resting RV isovolumetric pressure-measured Ees and model-estimated Ees (mean difference, −0.15 mmHg/mL [95% CI, −0.34 to +0.04]; $P=0.12$), and resting Ees values obtained by both methods were significantly correlated ($r=+0.48$ [95% CI, +0.13 to +0.72]; $P=0.01$; Figure S3).

When evaluating combined rest, 20-W, and peak exercise stages after sotatercept treatment, there was a consistent decrease in PA Ea when modeled across pooled rest and exercise stages (pooled mean difference across stages, −0.23 mmHg/mL [95% CI, −0.30 to −0.16]; $P<0.0001$), which was coupled with a similar decrease in model-estimated RV Ees (pooled mean difference across stages, −0.29 mmHg/mL [95% CI, −0.42 to −0.17]; $P<0.0001$), such that model-estimated Ea/Ees was unchanged (pooled mean difference across stages, +0.02 [95% CI, −0.09 to +0.12]; $P=0.74$; Figure 2 and Table 2). The decrease in Ees was related to the reduction in PA afterload as assessed by PA Ea ($r=+0.70$ [95% CI, +0.44 to +0.85]; $P<0.0001$) or pulmonary vascular resistance ($r=+0.68$ [95% CI, +0.41 to +0.84]; $P<0.0001$) (Figure 2). Because $dP/dt_{\max}$ is highly sensitive to loading conditions, we measured $dP/dt_{\max}$ adjusted for instantaneous pressure, which was not significantly altered at rest ($P=0.41$) after sotatercept treatment, consistent with the neutral effect on RV-PA coupling, and suggesting appropriate reduction in RV contractility relative to decreasing RV afterload. Even after adjustment for parenteral prostaglandin use, there remained a persistent reduction in RV Ees after sotatercept treatment (mean difference, −0.37 mmHg/mL [95% CI, −0.54 to −0.20]; $P<0.0001$) without interaction by parenteral prostaglandin use history ($P_{\text{interaction}}=0.16$).

Alternate measures of RV contractile performance by catheterization also decreased after sotatercept treatment, including pulmonary artery pulsatility index, RVSW, and RVSW/RVEDA (Table S1 and Figure 2). Measures of RV filling pressure and capacitance, including RVEDP and RVEDP/RVEDA, all improved after sotatercept treatment (Table 2).

Echocardiographic Measures of RV Systolic Performance

In contrast to measures of basal longitudinal motion (RV s' and TAPSE) that decreased at rest with sotatercept (Table 3), there was an improvement in global chamber contractile indices of FAC and RVFWS at rest. After sotatercept treatment, RVFWS increased across all rest and exercise stages (pooled mean difference across stages, +6.2% [95% CI, +4.5 to +7.9]; $P<0.0001$), indicating a redistribution of systolic shortening from the annulus to the RV free wall (Figure S4).

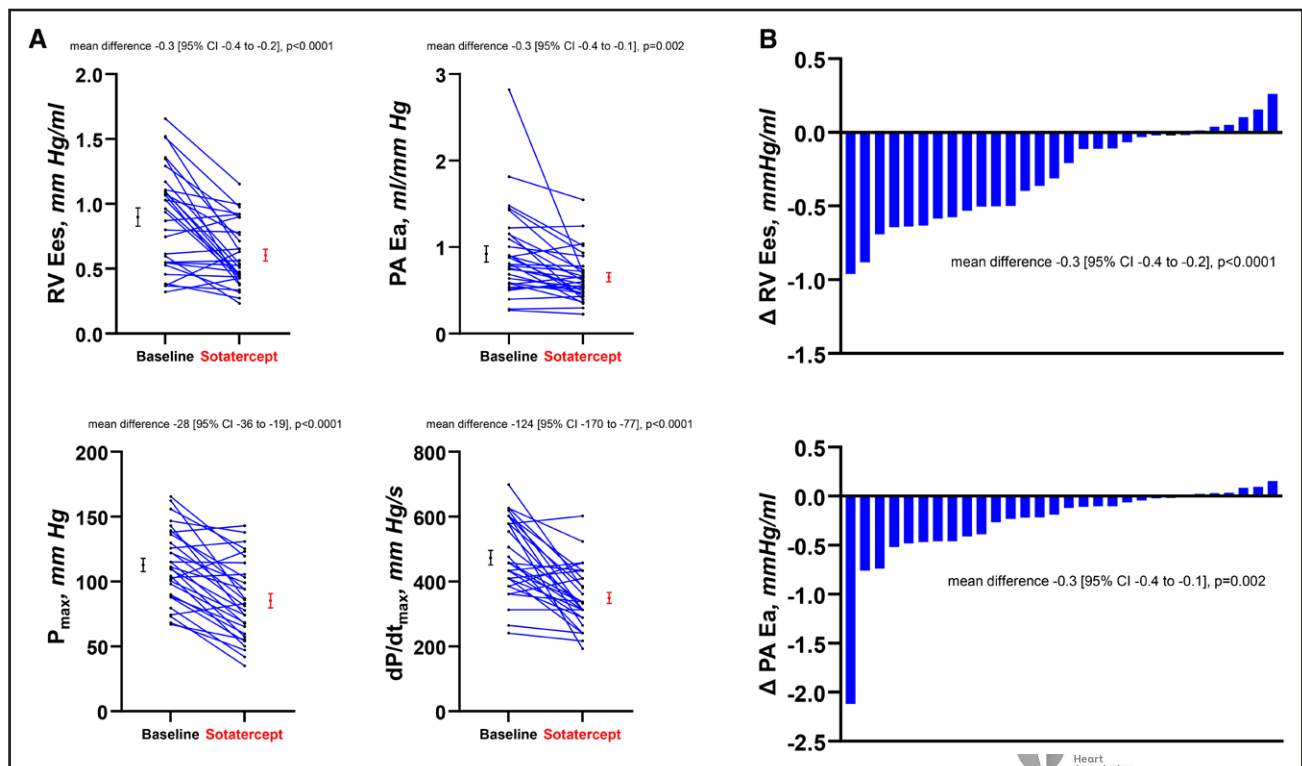


Figure 1. Effect of sotatercept on right ventricular contractility using the isovolumetric method.

Sotatercept decreased right ventricular (RV) contractility as assessed by end-systolic elastance (Ees), P_{max}, or dP/dt_{max} (A). There was also a corresponding reduction in pulmonary arterial elastance (PA Ea) that paralleled the reduction in Ees (B), suggesting an appropriate reduction in Ees relative to afterload.

Given the dependence of FAC on chamber remodeling, we explored relationships between RVEDA changes and FAC and RVFWS. The change in FAC after sotatercept treatment was no longer significant after adjustment for resting RVEDA in the mixed model ($P=0.35$), suggesting potential influence of decreasing RV size on the observed FAC increase after sotatercept treatment. In contrast, the change in RVFWS remained highly significant even after adjusting for differences in RVEDA in the mixed model ($P<0.00001$), suggesting changes in RVFWS were independent of decreasing RV size alone.

RA Remodeling, Function, and Pressure–Strain Relationship

Despite the observed decrease in RV contractility, there were favorable changes in RA size and function at rest and with exercise. RA strain was improved across all phases of the cardiac cycle, including reservoir strain, conduit strain, and booster strain during rest and exercise (Figure S4), with a decrease in maximal and minimal RA volume and improved RA ejection fraction and RA expansion index (Tables S2 and S3 and Figure S4).

Improvements in RA function were corroborated by favorable changes in corresponding RA pressure (Table S4), including a decrease in RA distending pressure (peak RA v wave–x wave nadir) and booster pressure (peak RA a

wave–y wave nadir) with improved RA capacitance and RA booster efficiency (Figure S5). There was also greater inspiratory decline in RA pressure after sotatercept treatment, with functional improvement in RA capacitance persisting not only at rest, but also with exertion (Tables S2 and S3).

DISCUSSION

The current data provide the first report of the effects of sotatercept on rest and exertional right heart mechanics in patients with PAH using standard measures including load-independent RV contractility and RV–PA coupling assessed by invasive catheterization. We show that RV contractility decreases after sotatercept treatment, which was corroborated by multiple independent measures (RV Ees, dP/dt_{max}, P_{max}, pulmonary artery pulsatility index, and RVSW) as well as clinically actionable measures of basal longitudinal RV function assessed by echocardiography. We observed that although sotatercept caused a reduction in RV contractility, the change in RV contractility was proportional to a reduction in RV afterload. These data emphasize that a reduction in RV contractility after sotatercept treatment should not universally be considered a maladaptive response, if it occurs in the context of a preserved RV–PA coupling relationship to maintain efficient right heart performance.²³ Despite decreases in RV Ees and longitudinal RV function (RV s' and TAPSE), global measures

Table 2. Effect of Sotatercept on Right Ventricular Contractility and Ventricular–Vascular Coupling During Rest and Exercise

Phase	Baseline	After sotatercept	Difference	P value	Difference across phases	Sotatercept P across rest and exercise	Phase* sotatercept P _{interaction}
RV–PA coupling							
RV Ees estimated, mmHg/mL							
Rest	1.03 (0.86 to 1.20)	0.64 (0.47 to 0.81)	−0.39 (−0.58 to −0.20)	0.0002	−0.29 (−0.42 to −0.17)	<0.0001	0.36
20 W	1.19 (0.94 to 1.43)	1.02 (0.77 to 1.26)	−0.17 (−0.43 to +0.09)	0.19			
Peak	1.30 (1.05 to 1.55)	0.99 (0.74 to 1.24)	−0.31 (−0.58 to −0.05)	0.022			
PA Ea, mmHg/mL							
Rest	0.92 (0.77 to 1.07)	0.65 (0.50 to 0.80)	−0.27 (−0.43 to −0.11)	0.002	−0.23 (−0.30 to −0.16)	<0.0001	0.56
20 W	0.95 (0.81 to 1.09)	0.70 (0.56 to 0.85)	−0.25 (−0.39 to −0.10)	0.002			
Peak	1.08 (0.92 to 1.24)	0.90 (0.74 to 1.06)	−0.18 (−0.33 to −0.02)	0.026			
RV waveform analyses							
RV Ees iso, mmHg/mL (rest)	0.90 (0.78 to 1.02)	0.60 (0.49 to 0.72)	−0.29 (−0.42 to −0.17)	<0.0001	–	–	–
RV Ea/Ees iso (rest)	1.06 (0.91 to 1.20)	1.13 (0.99 to 1.28)	+0.08 (−0.11 to +0.26)	0.42	–	–	–
Pmax, mmHg (rest)	112.9 (102.2 to 123.5)	85.2 (74.5 to 95.9)	−27.7 (−36.3 to −19.0)	<0.0001	–	–	–
RV Ea/Ees iso (rest)	1.06 (0.91 to 1.20)	1.13 (0.99 to 1.28)	+0.08 (−0.11 to +0.26)	0.42	–	–	–
dP/dt _{max} , mmHg/s (rest)	471 (431 to 511)	347 (307 to 387)	−124 (−170 to −77)	<0.0001	–	–	–
dP/dt _{max} /inst P, s ^{−1} (rest)	14.4 (13.0 to 15.8)	14.4 (13.0 to 15.8)	−0.0 (−2.1 to +2.0)	0.96	–	–	–
RVEDP, mmHg (rest)	15.6 (13.6 to 17.7)	11.0 (8.9 to 13.1)	−4.6 (−6.6 to −2.7)	<0.0001	–	–	–
RVEDP/RVEDA, mmHg/cm ² (rest)	0.49 (0.42 to 0.56)	0.39 (0.33 to 0.46)	−0.10 (−0.17 to −0.03)	0.007	–	–	–

Values represent marginal means (95% CI) from mixed models. Individual mixed models were used to compute marginal means and mean difference after sotatercept treatment at rest, 20 W, and peak exercise phases (left side of table). A separate mixed model was used to model pooled marginal mean differences across all phases after sotatercept treatment (right side of the table). dP/dt indicates change in pressure over time; Ea, effective arterial elastance; Ees, end-systolic elastance; inst P, instantaneous pressure; iso, isovolumetric pressure method; PA, pulmonary artery; RV, right ventricular; RVEDA, right ventricular end-diastolic area; and RVEDP, right ventricular end-diastolic pressure.

of RV and RA deformation improved, with enhanced right heart operating compliance and cardiac output reserve.²⁰ This observation suggests that changes in RV annular displacement, RV Ees, and Ea/Ees might be considered poor surrogates to understand the salutary effect of sotatercept on the right heart, while also pointing to a redistribution of RV systolic mechanics from the annulus to the free wall with sotatercept-related unloading. Our data suggest that measures of global RV performance, including the free wall, such as RVFWS or FAC, or integrated measures of upstream RA remodeling or function, such as RA strain, might provide more clinically useful measures to therapeutically monitor the right heart response to sotatercept.

RV Contractility Response With Sotatercept Compared With Vasodilators in PAH

Patients with PAH have a primary disease process in distal pulmonary arterioles that causes adverse changes in

the right ventricle as a secondary, pathogenic response. With gradual development of pulmonary hypertension, the right ventricle develops hypertrophy, with increased contractility as an adaptation to maintain RV–PA coupling. RV Ees has not been reported to decrease after vasodilator therapy, which primarily works by promoting vasodilation and a decrease in Ea to restore ventricular–vascular coupling.^{1,24,25} The preservation of Ees with pulmonary vasodilators may reflect, in part, their undesired systemic vasodilatory effects that may activate arterial baroreceptors with sympathetic activation, potentially mitigating expected declines in Ees from pulmonary vasodilation and decrease in Ea.⁵ Therefore, although PA vasodilators can improve RV–PA coupling, this may occur at the expense of an increase in inotropic recruitment and RV work as cardiac output necessarily increases to preserve systemic blood pressure.¹ In accordance, conventional measures of RV function by echocardiography, including TAPSE and RV s', generally increase after



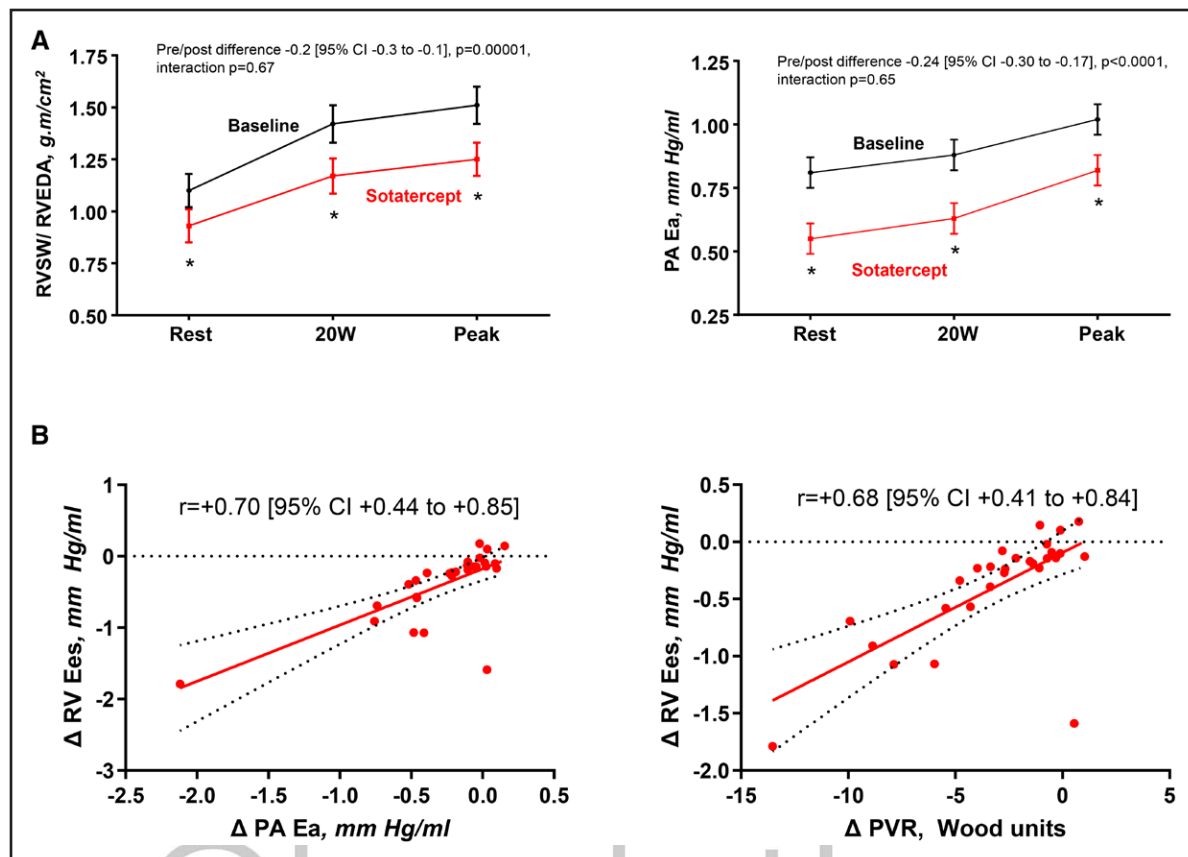


Figure 2. Effect of sotatercept on right ventricular contractility during exercise.

Sotatercept consistently decreased right ventricular (RV) contractile performance as assessed by RV stroke work (RVSW)/RV end-diastolic area (RVEDA) across rest and exercise, coupled with a proportional reduction in pulmonary arterial elastance (PA Ea; **A**). The reduction in RV afterload as assessed by PA Ea and pulmonary vascular resistance (PVR) was correlated to the reduction in RV end-systolic elastance (Ees; **B**).

vasodilator therapy, reflecting the improved ventricular-vascular coupling and increased RV work required for generating higher flow and cardiac output.^{26,27}

In contrast to vasodilator therapy, sotatercept markedly decreases pulmonary vascular resistance through nonvasodilatory mechanisms, without increasing cardiac output, and presumably without increasing sympathetic activation, when considering the reduction in heart rate observed in our cohort and others.^{5,20,28} This isolated pulmonary vascular effect might then be expected to manifest as an appropriate reduction in RV Ees driven by reversal of the Anrep effect (enhanced contractility to match an elevated afterload), with favorable RV and RA reverse remodeling, and overall decrease in RV work. This pattern of favorable decrease in Ees from nonvasodilatory alleviation in afterload is similar to the RV response from mechanical afterload reduction in chronic thromboembolic pulmonary hypertension²⁹ or even the LV response from mechanical afterload reduction with alcohol septal ablation in hypertrophic cardiomyopathy.^{30,31} The comprehensive exercise reserve characterization in the current study confirms that despite a decrease in Ees, right heart capacitance, free wall strain, and atrial function are all independently improved. Collectively, these data suggest

that decrease in RV Ees with sotatercept may represent a favorable adaptation to the nonvasodilatory decrease in afterload, restoring physiology closer to normal than can be achieved by vasodilator therapy alone. This is in striking contrast to a decrease in RV Ees that might occur in the setting of sustained pulmonary hypertension, which would be associated with RV-PA uncoupling, worsening right heart failure, and systemic venous congestion.

Clinical Implications

The confirmation of a physiologically appropriate decrease in RV contractility with preserved ventricular-vascular coupling after sotatercept treatment is a clinically relevant finding with important implications for noninvasive monitoring. It is common practice to noninvasively follow RV systolic performance in PAH using measurements performed at the base of the right ventricle, such as TAPSE and RV s'. However, in the phase 3 STELLAR (A Study of Sotatercept for the Treatment of Pulmonary Arterial Hypertension [MK-7962-003/A011-11]), these measures did not improve, despite substantial clinical improvement.³ In the current study, we further observed a decrease in TAPSE and s' at rest, potentially reflecting

Table 3. Effect of Sotatercept on Echocardiographic and Catheterization-Based Right Ventricular Performance During Rest and Exercise

RV function by echo and phase	Baseline	After sotatercept	Difference	P value	Difference across phases	Sotatercept P across rest and exercise	Phase* sotatercept P _{interaction}
RV free wall strain, %							
Rest	21.5 (19.3 to 23.6)	27.1 (25.1 to 29.1)	+5.6 (+2.7 to +8.5)	0.0004	+6.2 (+4.5 to +7.8)	<0.0001	0.10
20 W	24.1 (21.8 to 26.5)	28.5 (26.3 to 30.8)	+4.4 (+1.5 to +7.3)	0.004			
Peak	22.7 (19.9 to 25.4)	31.0 (28.6 to 33.5)	+8.4 (+4.7 to +12.1)	0.0001			
TAPSE, mm							
Rest	20.0 (18.4 to 21.5)	17.3 (15.7 to 18.9)	−2.6 (−4.3 to −1.0)	0.003	−0.9 (−1.9 to +0.1)	0.075	0.028
20 W	19.7 (17.8 to 21.6)	19.3 (17.3 to 21.2)	−0.4 (−2.6 to +1.7)	0.67			
Peak	20.4 (18.3 to 22.6)	21.1 (18.9 to 23.2)	+0.7 (−1.8 to +3.1)	0.57			
RV s', cm/s							
Rest	10.6 (9.8 to 11.4)	9.8 (9.0 to 10.6)	−0.9 (−1.6 to −0.1)	0.03	−0.6 (−1.1 to −0.0)	0.04	0.42
20 W	11.7 (10.6 to 12.8)	10.9 (9.8 to 12.0)	−0.8 (−1.6 to +0.0)	0.06			
Peak	12.5 (11.2 to 13.7)	12.5 (11.2 to 13.7)	+0.0 (−1.1 to +1.1)	0.99			
FAC, %							
Rest	28.3 (24.5 to 32.2)	33.9 (30.2 to 37.7)	+5.6 (+2.2 to +8.9)	0.002	+4.8 (+2.7 to +7.0)	0.0001	0.47
20 W	28.2 (24.3 to 32.2)	34.8 (31.1 to 38.6)	+6.6 (+2.5 to +10.8)	0.003			
Peak	30.7 (26.5 to 34.8)	33.5 (29.7 to 37.2)	+2.8 (−1.7 to +7.3)	0.21			

Values represent marginal means (95% CI) from mixed models. Individual mixed models were used to compute marginal means and mean difference after sotatercept treatment at rest, 20 W, and peak exercise phases (left side of table). A separate mixed model was used to model pooled marginal mean differences across all phases after sotatercept treatment (right side of the table). FAC indicates fractional area change; RV, right ventricular; and TAPSE, tricuspid annular plane systolic excursion.

the true reduction in RV contractility we identified using load-independent measurement of RV Ees. Therefore, TAPSE and s' may not be useful measures to clinically follow after sotatercept treatment initiation. In contrast, FAC and particularly RVFWS provide more integrated measures of global chamber performance including the free wall, and appear to more reliably capture salutary effects of sotatercept on RV mechanics after reverse remodeling. The discordant effects on RV annular and free wall function with sotatercept is interesting and may have bearing for other novel activin-targeting therapies that affect PA remodeling. The observation that RA strain and volumes also improve after sotatercept treatment may provide another reliable integrated measure of salutary right heart effects independent of the physiologically appropriate reduction in RV contractility.

Importantly, sotatercept does not appear to change RV–PA coupling as assessed using standard Ea/Ees assessment. It is common practice to use simplified indices of echocardiographic RV function indexed to PA systolic pressure (such as TAPSE/PASP) to estimate RV–PA coupling. Our data raise caution with this approach in sotatercept-treated patients, as we and others have reported favorable increases in such surrogate measures of RV–PA coupling with sotatercept (such as TAPSE/PASP),^{3,20} although no change in Ea/Ees could be detected after sotatercept treatment in the current study. This likely reflects the strong mathematical dependency of TAPSE/

PASP on PAP reduction, where favorable increase in this surrogate index may largely represent reduction in PAP after sotatercept treatment, without meaningfully informing ventricular–vascular coupling.

Limitations

Although the current analysis represents the first comprehensive report of changes in RV contractility indices after sotatercept treatment along with measures of RV–PA coupling, there are limitations to consider. Resting Ees was measured directly using high-fidelity RV pressure waveforms, but exercise Ees was estimated using an iterative model-based approach. Although we validated model-estimated Ees against measured Ees at rest, and the model-based approach implicitly derives biventricular Ees in its iterative computational approach,¹⁷ the model-based estimates of exercise RV Ees were not independently validated against exercise pressure-based methods, requiring further study.

CONCLUSIONS

Sotatercept decreases load-independent measures of RV contractility in proportion to reduction in afterload, with preserved ventricular–vascular coupling, reflecting a physiologic adaptation to the decrease in RV afterload. Despite this reduction in contractility, RV and RA

mechanics improve after sotatercept treatment, with enhanced exertional cardiac reserve. These data have implications for optimal monitoring of patients after sotatercept treatment, in whom measures of global right heart remodeling and strain improve, but these changes may be disconnected from conventional measures of basal longitudinal assessment of systolic performance, such as RV s' or TAPSE, which change in a directionally distinct pattern from measures of deformation.

ARTICLE INFORMATION

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Supplemental Material

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